Thomas Jefferson **Physics Olympiad**

A high school physics contest

TJ Physics Team <tjhsstphysicsteam@gmail.com>

The James Webb Space Telescope

If you have any questions or concerns, please email TJ Physics Team.

Contents

Rules

Teams will have from 9:00 PM EST on March 13th to 9:00 PM EST on Marth 15th to complete the TJPhO. All teams are required to submit their response with a cover page listing the team name and number, team member names, team email, and the date. Each submitted page should also have the problem number. All other formatting decisions are delegated to the teams themselves, with no one style favored over another. We suggest that teams use LaTeX for typesetting. The easiest way we've found to typeset LaTeX is with Overleaf.

Problems

Many of our problems will focus on the James Webb Space Telescope and the foundational physics that went into designing it. There is also a wild card section which has questions on an assortment of different topics – it is up to you to decide what approach you need to use. Lastly, we have a coding section, where in addition to doing some math you are expected to write a program in a language of your choice and submit your code for us to review.

Our problems were written by Tarushii Goel, Cyril Sharma, Anya Mischel, and Alvan Caleb Arlandu. We would like to thank Dr. Jonathan Osborne for reviewing our problems and providing valuable feedback.

Collaboration

Students participating in the competition may only correspond with other members of their team. No other correspondence is allowed, including mentors, teachers, professors, and other students. Teams are not allowed to use online/print resources or post content on online forums asking questions related to the exam. We welcome teams to email us if there are any questions or concerns. If a problem explicitly states that it is allowed, teams may use computational resources they might find helpful, such as Wolfram Alpha/Mathematica, Matlab, Excel, or programming languages.

Submission

Teams must submit their solutions by email to tjhsstphysicsteam@gmail.com by the deadline 9:00 PM EST on May 15th. Late submissions will not be considered. Solutions should be written in English and submitted as a single PDF document with the .pdf extension. The email must contain "Submission" in the subject line. Only one person per team, the person who registered through the Google Form, should send the final submission.

Awards

The top five teams will receive awards.

1st place: $200, 3 one-year subscriptions to WolframAlpha Notebook Edition

2nd place: $150, 3 one-year subscriptions to WolframAlpha Notebook Edition

3rd place: $100, 3 one-year subscriptions to WolframAlpha Notebook Edition

4/5th place: 3 one-year subscriptions to WolframAlpha Notebook Edition

Sponsors

We are incredibly grateful to our sponsors, Dr. Adam Smith and Mr. Robert Culbertson, and Thomas Jefferson High School for Science and Technology for their support. We would also like thank our sponsors WolframAlpha and the Thomas Jefferson Partnership Fund for making this Olympiad possible.

Introduction

The James Webb Space Telescope (JWST) was developed in a collaboration between NASA, the European Space Agency (ESA), and the Canadian Space Agency. It is named after James E. Webb, who was the administrator of NASA from 1961 to 1968. Development began in 1996 on a $500 million budget with a launch initially planned for 2007. Instead it was launched last year and costed an esitmated total of $9.7 billion.

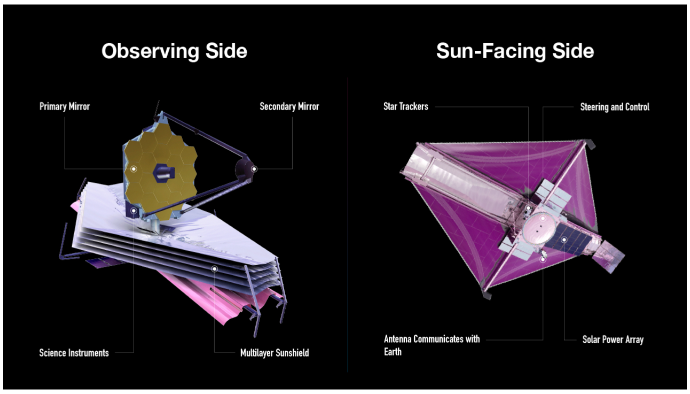

Figure 1: The James Webb Space Telescope

It is designed to perform near-infrared astronomy and is capable of seeing objects that are too faint and distant for the Hubble Space Telescope to capture images of. It can see as far back as 50 million years after the Big Bang.

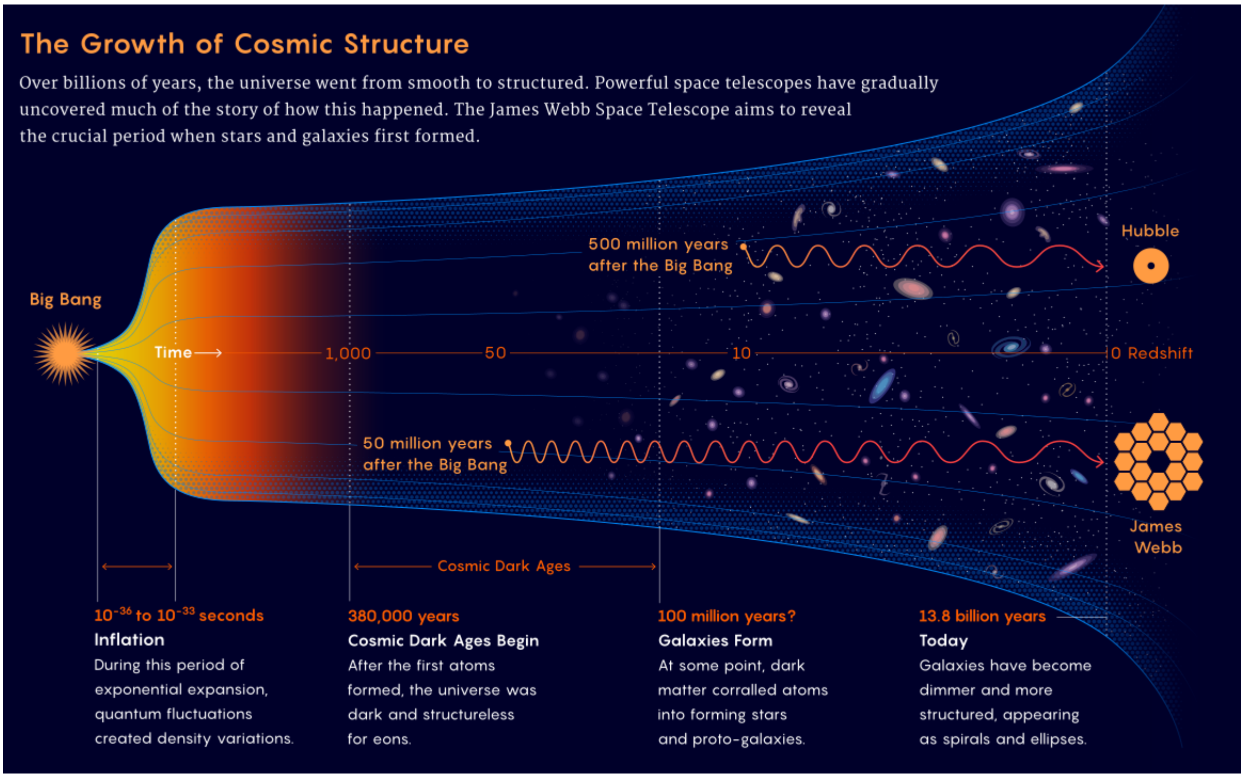

Figure 2: History of the Universe

The JWST was launched along with the Ariane 5 rocket on December 25th, 2021, from a ESA launch site in French Guiana. On December 27th, it sailed beyond the orbit of the moon, and on December 31st, it began the process of unfurling its massive sun shield and tightening the layers, a process that took until January 3rd, 2022. On January 24th, it finally arrived at the second lagrange point, L2, where it will remain while it takes high resolution images of space (previously, the Herschel Space Telescope and Planck Space Observatory have also orbited at the L2 point). The JWST has finished unfolding its mirrors and has spent the last few months aligning its optical instruments. Just a few days ago, it reached the end of its alignment process and took a high resolution image, which is included in Figure 4.

Figure 3: The first image taken by the JWST, at the beginning of the long process of alignment. THe light is from a star named HD84406. Each of the 18 bright dots is the light reflected by one of the JWST's 18 primary mirror segments. (Image credit: NASA/ESA/CSA/STScI)

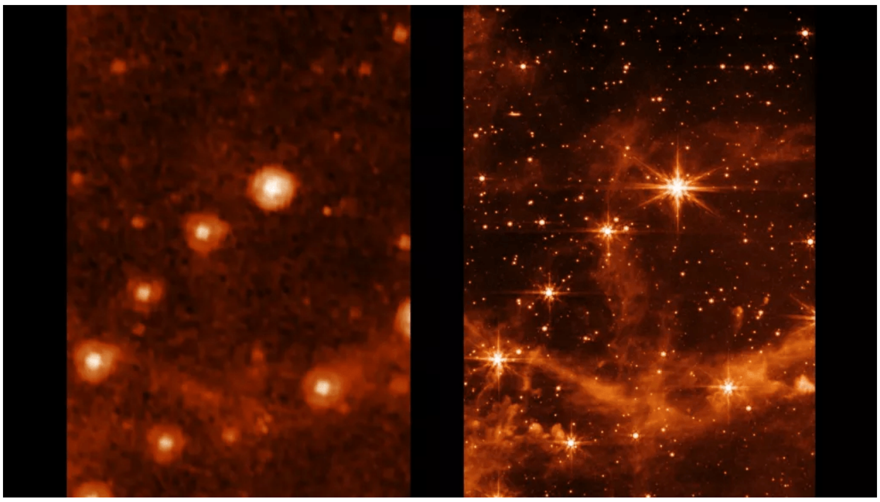

Figure 4: The Large Magellanic Cloud, as seen by NASA's Spitzer Space Telescope (left) and the JWST after a long period of alignement (right). (Image credit: NASA/JPL-Caltech (left), NASA/ESA/CSA/STScI (right))

The James Webb Space Telescope (JWST)

1. *Lagrange Points.* When communicating with a satellite, it's desirable if the satellite is always in the same place. Lagrange points are thus an attractive destination for satellites; as they remain stationary relative to the Earth (they orbit the sun with the same angular velocity as Earth). For this reason, the JWST has been sent to orbit the second Lagrange point, shown below.

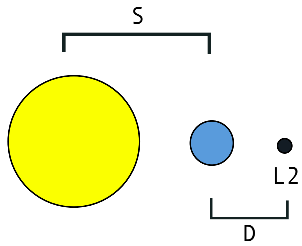

Figure 5: Langrange Point

   (a) (1 point) Calculate the distance to L2, D, in terms of the mass of the earth and sun, M_e and M_s, and the distance of the sun from the earth, S. Hint: L2 does not move relative to the Earth, meaning it orbits the sun with the same angular velocity as the Earth.

   (b) (1 point) Calculate the gravitational potential energy as a function of the distance from the sun in the direction of the earth. You may assume the Earth and Sun are point masses. The JWST has mass m_j.

   (c) (1 point) The potential function calculated above doesn't account for the centripetal force needed to stay in the same relative position. Pick a frame of reference that rotates around the sun with the earth. From this frame of reference, a centrifugal force acts on all bodies away from the axis of rotation. Compute the effective potential function, a potential function that takes into account both the centrifugal force and the gravitational forces.

   (d) (1 point) Contrary to the calculated potential function, L2 is actually fairly easy to orbit. This is because the Coriolis force helps stabilize L2 orbits. The diagram below shows how the Coriolis force and gravitational forces balance, allowing an object to stay in a stable orbit around L2. Qualitatively explain how an object moved slightly towards the Earth from its spot on L2 would find itself at this steady state in terms of the Coriolis and gravitational forces.

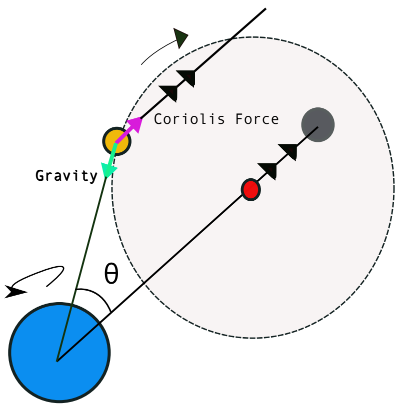

Figure 6: Stable Orbit of Lagrange Point

(e) (1 point) Let the distance between the Earth and the Sun be S. Let the distance along the x axis from the Earth to the JWST be D. You may assume the sun's gravitational pull is not significantly angled, and is directed purely in x direction, as shown in the diagram. In terms of S and D, what angle θ must the JWST be at relative to the Earth in order to remain stationary relative to the Earth?

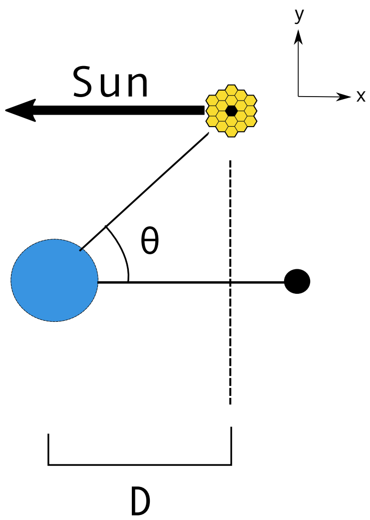

Figure 7

2. *Solar Radiation Effects.* The JWST's sensitive infrared imaging technology must be

kept at low temperatures to operate correctly. For this reason, the JWST has a sun shield made of Kapton E that blocks solar radiation. We can model the sun shield as a flat surface being struck by electromagnetic (EM) waves.

The energy flux of an EM wave is known as the Poynting vector, $\mathbf{S}$. This fluctuates with the frequency of the EM wave, but generally all we care about is the average of the magnitude of the Poynting vector, $\langle S \rangle$, which is also known as the light irradiance, I. The momentum density of EM waves is $\mathbf{P} = \frac{\mathbf{S}}{c^2}$. For this problem, assume that the Poynting vector is perpendicular to the shield.

(a) (1 point) Express the average pressure p_a exerted on the sun shield as a function of I and c, assuming the shield absorbs 100% of the radiation. Then calculate the pressure p_r when it perfectly reflects the radiation.

(b) (1 point) The shield is 22 by 10 m in size, its reflectance factor is 0.8, and the irradiance of light at the L_2 point, where the JWST is situated, is about 1366 W/m^2 (this is actually the irradiance of light at the Earth, but the L2 point is only 0.01 AU away so the difference is negligible). Calculate the total force solar radiation exerts on the shield.

(c) (2 points) Random fluctuations in solar irradiance can cause pressure gradients across the sun shield, which result in a torque. Use the set up shown below, where the force calculated in the previous question is directed towards the very end of the shield rather than evenly distributed, to calculate the torque on the shield. The effects of this torque need to be mitigated in order to keep the shield facing the sun and protect the JWST's sensitive equipment. One approach of doing so is to use a momentum wheel with stored angular momentum. Using the set up shown below, where the torque you calculated is exerted on the momentum wheel for a period of one day, to find the total rotation θ in terms of the angular momentum L of the momentum wheel, which is directed downwards on the figure below ($I_W >> I_S$, where I_W and I_S are of moments of inertia about the COM of the wheel and shield for the wheel and shield, respectively). Graph θ vs. L. What is the axis of rotation?

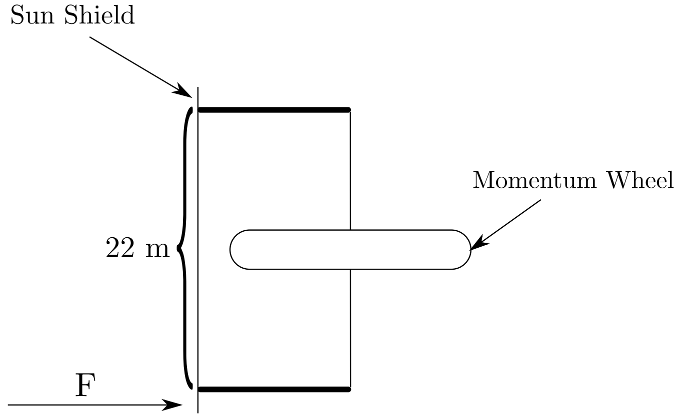

Figure 8: Top View of Shield-Wheel Configuration

3. (5 points) *Exoplanets.* Suppose the JWST discovers a faraway exoplanet orbiting a star in an elliptical orbit with eccentricity 0.42 and a semi-major axis of 1.9 AU. In order for this particular planet to have the possibility of carbon-based life, the

planet spend at least 20% of the time of its orbit within 1.4 AU of the star. Given this information, does the planet stay within the maximum habitability distance long enough to have the possibility to support life? *Wolfram Alpha or other integration-solving software is permitted for this problem only, but all steps including the integral expression must be explicitly written.*

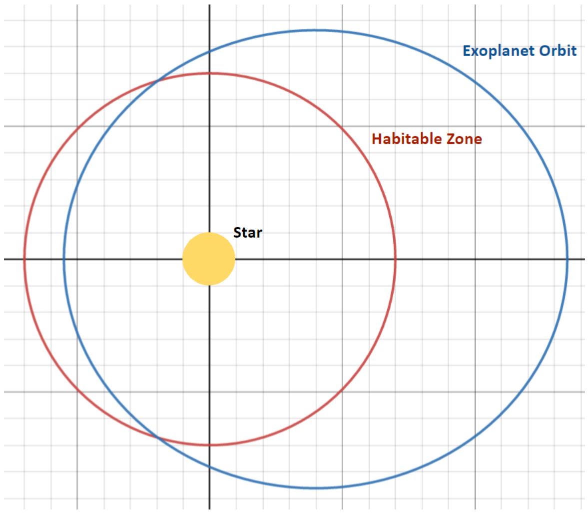

Figure 9: Exoplanet Orbit and Habitability Boundary

Imaging in the James Webb Space Telescope

4. *Lenses.* Lenses rely on the law of refraction, which states that light will bend when transitioning between two media. Specifically, $n_1 \sin\theta_1 = n_2 \sin\theta_2$, where n_1 and n_2 are the indices of refraction of the materials (phase speed $v = c/n$), and θ_1 and θ_2 are the angles that the incident and transmitted light rays make with the perpendicular to the boundary between the materials. JWST's lenses use refraction to concentrate light onto a focal point, thereby creating high resolution images.

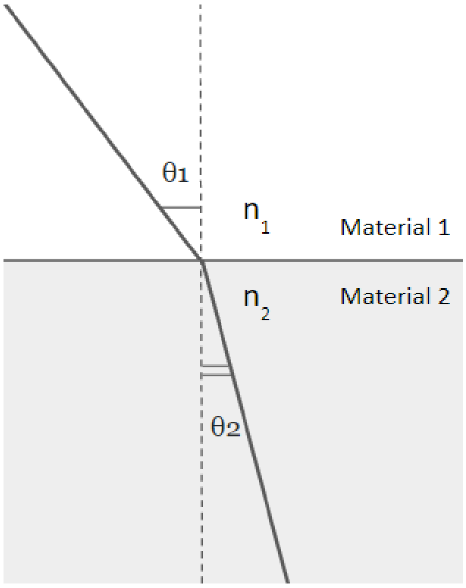

Figure 10: Refraction of Light Passing Through Different Materials

(a) (3 points) Suppose you have a lens as shown in the setup below (gray region is glass), which focuses light from point source S in space at focal point P inside the lens. According to Fermat's principle of least time, both light rays shown must take the same amount of time to travel between the two points. By relating the travel times of the two rays, establish a relationship between s_1 and s_2 in terms of R, the radius of the lens, n_1, and n_2. Assume $d_1^2 - s_1^2 = (d_1 - s_1)2s_1$, $d_2^2 - s_2^2 = (d_2 - s_2)2s_2$, and $R^2 - x^2 = (R - x)2R$.

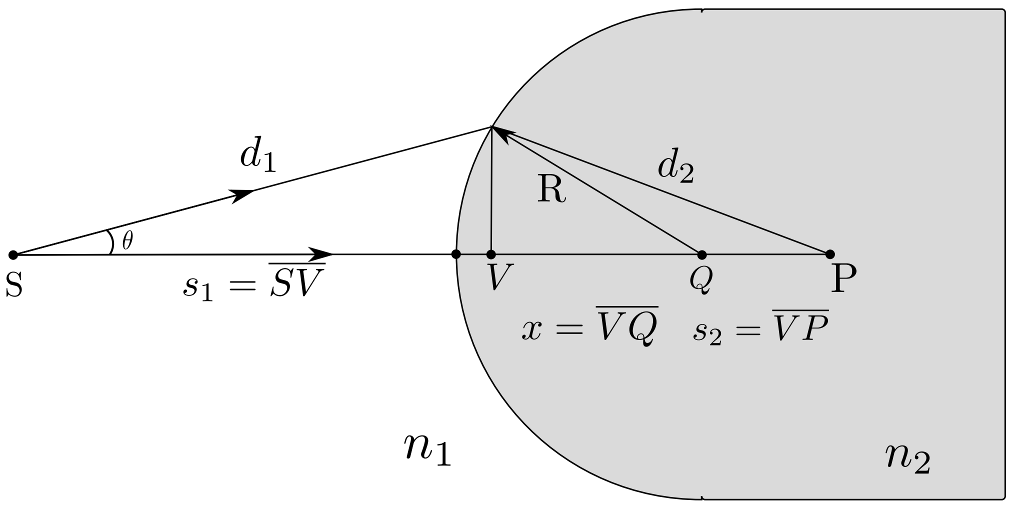

Figure 11: A Lens

(b) (2 points) Replace the lens used in the previous question with a lens of the shape shown in the figure below, and derive a new expression relating s_1 and s_2, in terms of R_1, R_2, n_1, and n_2. Assume $d << s_1, s_2$.

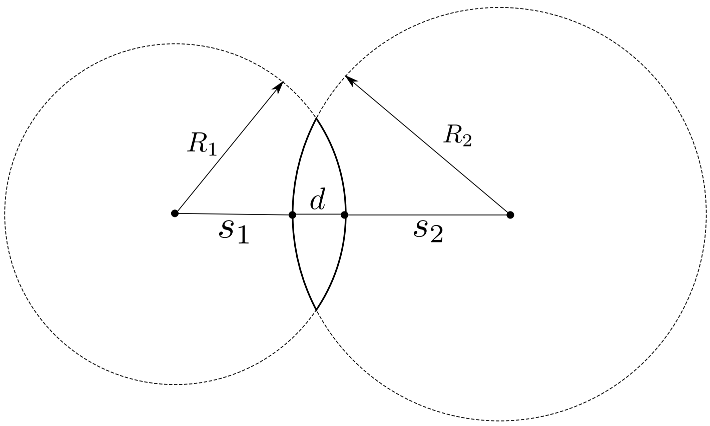

Figure 12: A Thin Lens

(c) (1 point) Generally, the sources of light which a lens is engineered to focus are much farther away than from the lens than the focal points of those sources of light (e.g. the JWST images stars many light years away but the star image may be constructed a few centimeters away). The distance at which the image focuses is the f, the *focal length*, which is derived by letting $s_1 = \infty$. Plug this value for s_1 into you answer for the previous part to find the focal length.

(d) (1 point) Light waves warp while traveling through space, and being able to measure these distortions and adapt optical systems accordingly is crucial. One method of measuring wavefront distortions is to arrange lenslets in a line, as shown in the figure below. Each lenslet will focus light on a different point on the image plane (see figure below), due to the difference in the angle at which the wavefront strikes the lens. Demonstrate how the lenslet array distinguishes between different wavefronts by marking the focal point of each lenslet for a planar wavefront. Then mark the focal points for the distorted wavefront, which may be shifted up or down. You should draw a total of 10 points, 5 on each image plane.

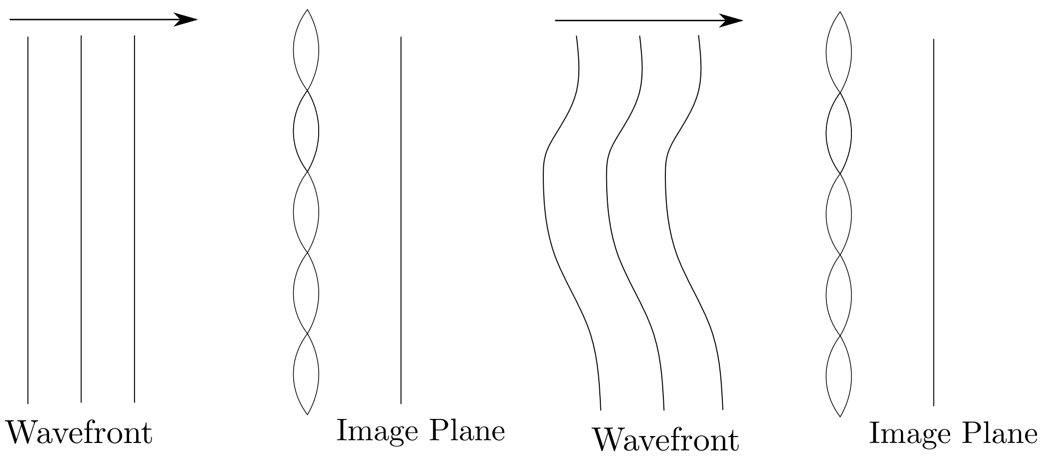

Figure 13: A Lenslet Array

5. *Diffraction.* The JWST's optical equipment is precisely fine-tuned so as to produce the highest resolution images possible. However, the resolution of these images has a theoretical limit due to the diffraction of light.

(a) (3 points) Let's say you have a line of n harmonic electron oscillators spaced by a distance d that are moving in phase. The electric field created by single oscillator at a *large distance* r away from can be approximated as $A\cos{(wt - \frac{2\pi r}{\lambda})}$, where λ is the wavelength of electromagnetic wave, and w is its angular velocity. Write the resultant amplitude of the field oscillation A_R at a point a distance r away and at an angle θ (in terms of A, f, the frequency of oscillation, d, and θ). Note that in the diagram below the light rays reaching the point are shown as approximately parallel (and you can assume they are) because the point is very far away.

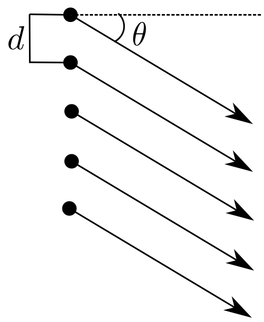

Figure 14: Electron Oscillator Setup where $n = 5$

(b) (1 point) Plot I (the light irradiance, which is explained in the solar radiation problem) vs. θ. At what values of θ do you see the greatest irradiance of light? Where is the irradiance 0?

(c) (1 point) Let's think of a one of the JWST's lenses as a small, circular opening in a barrier that allows light to pass through. The theory of diffraction states that the light passing through the opening is equivalent to that which would be generated by a lot of (really, infinitely many) sources of light of the same wavelength evenly separated by a small distance across the opening. So, the light that passes through the opening forms a diffraction pattern that is quite similar to that which you calculated in previous parts. Draw the diffraction pattern you would expect (it does not have to be mathematically precise, we are only looking for the general position of high irradiance and low irradiance regions).

Interestingly, the angle θ at which the first minimum in light irradiance occurs, measured from the direction of incoming light, is approximated by the formula $\sin\theta = 1.22\frac{\lambda}{d}$ where d is the diameter of the aperture. So, using the Raleigh Criterion, θ is the smallest angular separation you can have between two objects while still being able to resolve them. This is what we mean when we say that the resolution of a camera is *diffraction-limited.*

Additional Assorted Problems

6. *Windmills.*

(a) (1 point) Windmills generate energy by slowing down the wind and extracting its energy. Suppose we have a windmill as shown in the figure below. Assuming that we are dealing with incompressible, irrotational, and steady flow of air, which of the colored lines (red, blue, or black) best represents the general shape of the streamtube of air that passes through the windmill?

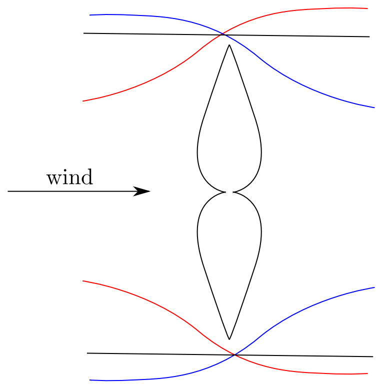

Figure 15: Top View of a Windmill

(b) (2 points) Bernoulli's equation, which is derived from conservation of energy, states that $P + \frac{1}{2}\rho v^2 + \rho g h$ is remains constant in steady flow. Assuming that energy is not inputted or outputted into the streamtube upstream or downstream of the windmill (only at the windmill), apply Bernoulli's equation to find the force exerted on the windmill. Express you answer in terms of W, the undisturbed wind speed, A_w, the area of the windmill, ρ, the density of air, and a, the induction factor, which is related to the design of the windmill and indicates much energy it extracts (specifically, the speed of air at the windmill is $W(1-a)$), and the unknown W_∞, which is the speed of the air far downstream. Assume that gravitational potential energy is unchanging during fluid flow. Additionally, draw graphs of the pressure and wind speed vs. position along the streamtube (these only need to be correct to the first derivative).

(c) (1 point) Derive another expression for the force on the windmill, this time applying conservation of momentum (in terms of the same variables as the previous part).

(d) (1 point) Combine the expressions you found in the last two parts to find W_∞ in terms of W

(e) (1 point) Plug your expression for W_∞ back in to get a formulation for power extracted (assume 100% efficiency). What induction factor gives you the highest power? What is the ratio between the maximum power you can extract and the power of the wind passing through the windmill?

(f) (1 point) Now let's consider a contraption composed of two propellers, one in the air and one in the water, connected by a shaft, as shown in the figure below. The analysis for the the force on power extracted by the propeller in the air is exactly the same as that established in the previous parts. However, the propeller in the water must be reanalyzed (though the steps to do so are very similar) because it functions more like a fan than a windmill, speeding up the water to a speed V at the propeller. Calculate the power delivered by the water propeller to the water in terms of ρ_2, A_2, V.

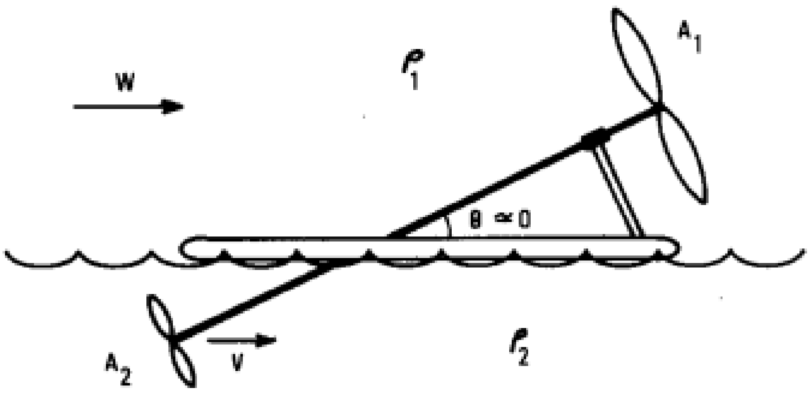

Figure 16: Propeller Contraption

(g) (1 point) Assume $P_{in} = P_{out}$, and calculate the net force on the contraption using the induction factor that provides maximum efficiency (calculated in a previous part). You answer should be in terms of a subset of these variables: $\rho_1, A_1, \rho_2, A_2, W$.

(h) (2 points) What is the maximum speed of this contraption, in terms of W? (the answer may be surprising)

7. *Ceaseless Cylinders.* Two infinitely long concentric cylindrical conducting shells are grounded (this means that they are connected to an infinite source/sink of charge, and charge can be induced on the conductors in order to maintain a potential of 0). The outermost conductor (conductor B) has a radius b, and the innermost conductor (conductor A) has a radius a. A single point charge q is place at some radius r ($a < r < b$), inducing charge on the conductors (Q_A and Q_B).

(a) (1 point) What is the total charge induced on both conductors, $Q_A + Q_B$? (in terms of some subset of the following variables: a, b, q, r). Provide a short explanation for how you can determine this without knowing Q_A or Q_B.

(b) (6 points) Green's reciprocity theorem states that $\int_{allspace} \rho_1 V_2 dV = \int_{allspace} \rho_2 V_1 dV$, where ρ_1 and V_1 are the charge density and potential in one charge configuration, ρ_2 and V_2 are the charge density and potential in a completely separate charge configuration, and dV is the differential volume. Use this theorem to find Q_A and Q_B individually. (in terms of some subset of the following variables: a, b, q, r).

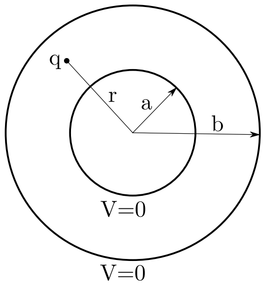

Figure 17: Sketch of the Situation

8. *Structures.* Stress represents the internal forces an object experiences, defined as the force over a cross-section divided by the area of that cross-section $\sigma = \frac{F}{A}$. Strain represents the elongation caused by a stress, defined as the change in length over the original length $\epsilon = \frac{\triangle l}{l}$.

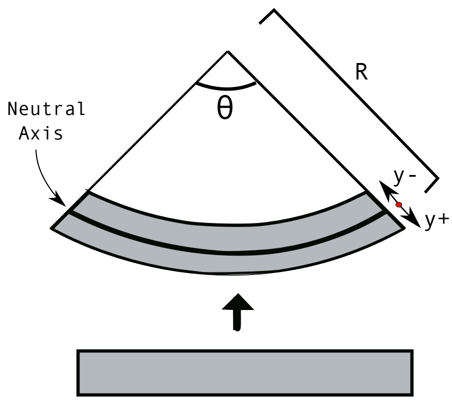

Figure 18: Bending Stress

(a) (1 point) In the diagram above, a beam is bent to have a radius of curvature R. Calculate the strain in the beam as a function of the displacement, y, from the neutral axis, where positive displacement takes you further from the center of curvature, as indicated in the diagram.

(b) (1 point) Stress is related to strain through Hooke's law, $\sigma = E\epsilon$, where E is Young's modulus and ϵ is the strain. Young's modulus is the property of the material the rod is made out of, and characterizes its response to stress. Using the above calculation for strain above, calculate the stress in the beam as a function of y, R, and E.

(c) (2 points) Using your expression for stress from part b, calculate the bending moment on the beam in terms of its moment of inertia I. Then plug your answer back in to your calculation of stress to determine a formula for stress independent of the radius of curvature.

(d) (2 points) The formula for stress you calculated above is used for calculating the bending stresses in a material. Another type of stress is known as direct stress, resulting from the application of a load (force over an area) which does not induce bending. In the diagram below, both bending stresses and direct stresses are present. Assuming the building has no strength under tension (an assumption that works well for structures made solely of concrete), what load displacement ϵ will cause the building to break under a load P?

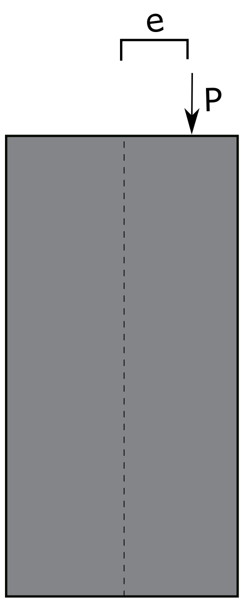

Figure 19: Building

9. *Intermediate Axis Theorem*

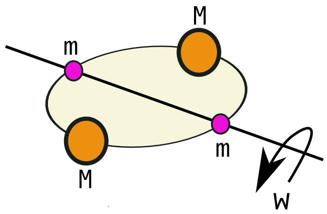

Figure 20

(a) (1 point) In the above diagram, a thin, rigid, massless disk spins along the axis shown. The orange masses (M) have substantially more mass than the purple masses (m). When viewed from a reference frame that rotates with the disk (at angular speed ω), what centrifugal force per unit mass would a test mass m_t experience as a function of their distance (r) from the axis of rotation.

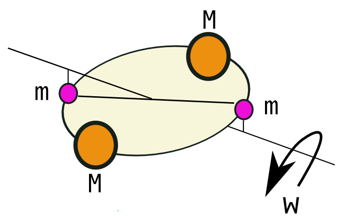

Figure 21

(b) (3 points) Suppose the disk of radius R is disturbed an infinitesimal amount mid-rotation as shown in the above diagram. The purple masses now feel a centrifugal force. Calculate a differential equation which describes the angle of the small masses (m) with respect to the center of the disk as a function of time. Qualitatively describe the solution to this differential equation. You may assume the disk is in space, so no gravitational force, and that the tension forces are insignificant to the large masses, meaning the large masses stay in essentially the same location. Furthermore, since the purple masses are so small, you may assume the Coriolis force is not sufficient to overcome tension forces and thus can be ignored.

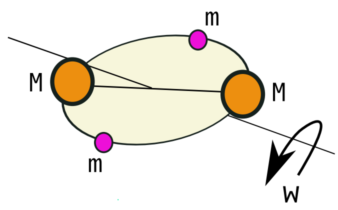

Figure 22

(c) (1 point) The Coriolis force can be calculated as $2m(\vec{\omega} \times \vec{v'})$, where m is the mass of the object experiencing the force, $\vec{w}$ is the vector describing the angular rotation of the reference frame, and v' is the velocity of the object experiencing the force in the reference frame. Now suppose that the masses are disturbed as shown in the above diagram. Qualitatively explain how the Coriolis effect will modify the analysis from the previous part.

Coding Problems

10. *Grover's Algorithm.*

A qubit is the quantum equivalent of a classical bit. Like a classical bit, qubits can only be measured in one of two states, denoted 0 and 1. Unlike a classical bit, the state of a qubit is not determined before the measurement. Before measurement, qubits exist in a state that is a superposition of the measureable states, which after a measurement, collapses problematically into one of the two measurable states.

A qubit's state is a vector of two complex numbers $\begin{bmatrix} a+bi \\ c+di \end{bmatrix}$. Each number in the state vector is associated with a specific state. Typically the first number is associated with $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and the second is associated with $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The probability of a qubit collapsing into a state is given by the magnitude squared of the associated component in the state, for example, if my state vector is $\begin{bmatrix} a+bi \\ c+di \end{bmatrix}$, then to calculate the chance the qubit will collapse into the state $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is $|a+bi|^2 = (\sqrt{a^2+b^2})^2 = a^2+b^2$. Since the qubit must collapse into one of the measurable states, the sum of the magnitudes squared (the sum of the probabilities) for all the qubit's state must add up to 1.

More generally, a quantum system can be represented as a vector of amplitudes that represent the probability of collapsing into a given state. For example, a quantum system of two qubits can be measured to be in one of four possible states: $\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\0\\1\\0\end{bmatrix}$, $\begin{bmatrix}0\\0\\0\\1\end{bmatrix}$. A possible representation for the state could be $\begin{bmatrix}0.5\\0.5\\0.5\\0.5\end{bmatrix}$. The magnitude squared denotes the probability of being in a given state, so the probability of being in state $\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$ is $|0.5|^2 = 0.25$. Note that the sum of the probabilities add to 1, $0.25*4 = 1$, as the qubit must collapse into some state.

Furthermore, if vectors are representation of quantum states, matrices are representations of quantum transformations, processes that convert one quantum state into another.

(a) (1 point) Suppose we have a quantum system with 2^N possible states. Let the initial state of the system be s the equal superposition state s. Consider the pure target state w and the equal superposition state s' that is orthogonal to w. If $s = c_1 w + c2s'$, find c_1 and c_2.

(b) (2 points) U_s, also known as the diffuser, is a transformation computed through the formula $U_s = 2ss^\top - I$, where s is the initial state and s^T is the transpose of s. Compute $U_s s$. Compare the effect of U_s on s with U_s on w where w is orthogonal to s.

(c) (1 point) Let U_f be the transformation which negates the amplitude of the target state. Compute $U_s U_f s$.

(d) (1 point) Each application of $U_f U_s$ induces a rotation on the input state, calculated above as θ. For what n is $\|(U_f U_s)^n s - w\|$ minimized? In other words, how many applications of $U_f U_s$ are necessary to rotate the input state, s as close to the target state, w as possible? Show your derivation.

(e) (3 points) **Code**: The procedure outlined in the previous steps, applying U_f and U_s repeatedly to a state are known as Grover's algorithm. Write a computer program that utilizes Grover's algorithm to find a provided target state, say $|100010\rangle$ given an arbitrary number of starting states. Specifically, write a function that takes in a target state and a value representing the total number of states. Your function should represent the operators U_s and U_f as matrices, and should output the probabilities of measuring the target state after the optimal number of iterations and the number of iterations your algorithm used.

11. *JWST Collision*

Suppose when the Earth is at the point closest to the Sun, the perihelion, the JWST, with a mass of 6161.4 kg, is stationed at the L2 point in stable orbit. Suppose a rogue satellite, of mass 2547.3 kg, collides with the JWST such that both satellites remain intact. At the point of instantaneous collision, the velocity of the asteroid is $3000m/s$, parallel to the velocity of the JWST which we consider to be in the $+\hat{\jmath}$

direction. For the below problems, use the standard numerical values for the mass of the Earth and Sun.

(a) (1 point) If the collision is perfectly inelastic and the asteroid's velocity is in the opposite direction of the JWST's, compute the velocity vectors of the JWST and the rogue asteroid immediately after the collision.

(b) (1 point) If the collision elastic and the asteroid's velocity is the in the same direction of the JWST's, compute the velocity vectors of the JWST and the rogue asteroid immediately after the collision.

(c) (2 points) The coefficient of restitution is defined as the ratio of the final to initial relative speeds between the two objects. If two objects with velocities u_a, u_b and masses m_a, m_b respectively collide with coefficient of restitution C_R, derive the final velocities v_a, v_b of both objects after impact.

(d) (4 points) **Code**: Assuming elastic collision, plot the trajectory of both the JWST and the rogue satellite through the first 30 days after collision. Provide the final positions of both satellites at the end of the time period. Explain with reasoning what type of equilibrium exists at the L2 point?

(e) (2 points) **Code**: Assuming elastic collision, Plot the distance of the JWST to the L2 point over the first year after collision. Provide the final position of the JWST at the end of the time period.

Additional Resources

Below is a list of additional information about the James Webb Space Telescope and its recent advances for those interested. Click on the hyperlink to visit the page.

Design of the JWST

- Optics on the JWST
- Full Summary and Detailed Design Overview
- Orbit of the JWST
- Operating Temperature

JWST Timeline

- Timeline of JWST